QP-based impact momentum maximization for a hammering task by a humanoid robot

Jimmy Vu†, Ruud Erens†, Hélène Stefanelli†, Rafael Cisneros-Limón†*, Mehdi Benallegue† and Abdelaziz Benallegue†#

***Abstract*—Humanoid robots can offload repetitive, load-bearing operations that expose workers to cumulative joint stress. In this paper, we focus on a hammering task with our humanoid robot HRP-5P and we propose a method to hit a nail with a given impact for a desired hitting velocity based on Quadratic Programming and the maximization of both the effective mass of the robot and impact momentum. Moreover, we implement a 5th order Bézier curve as the trajectory, which ensures smooth, jerk-bounded motions from any reachable initial pose.**

***Index terms*—** quadratic programming, hammering, humanoid robot, effective mass, end-effector trajectory

I. Introduction

Manual workers in physically difficult and wearing industries, such as the construction industry, have a 50% higher risk of developing musculoskeletal disorders than people in other industries [1]. Inadequate working posture and repetitive movements have been identified by Côté et al. [2] to contribute to musculoskeletal disorders, which is the case for a hammering task. Furthermore, they demonstrated that fatigue affects elbow motion and shoulder injury affects both wrist and elbow motions during hammering. Balendra et al. [3] showed that the increased moment exerted on joints during hammering can lead to more injuries, where a larger hammer causes a bigger increased moment. Unfortunately, larger hammers are often the preferred choice to accomplish such task. Indeed, as a result from Petrič et al. [4] indicates, hammers themselves are one of the non-powered handtools that cause the most injury [5], especially to the fingers which are the most injured body part. Another cause of potential physical damage to workers is the posture that the hammering task has to be carried out with: Yuan-Ho Chen and Yi-Lang Chen [6] studied the optimal height to realize this task and concluded that it should be between 25 cm and 35 cm below the elbow height. However, it is not always possible to be at this height due to the variable working area in construction.

To relieve these workers from such joint-stressing activites, AIST developed the HRP-5P humanoid robot [7] which can take care of the wearing and exhausting tasks, such as carrying heavy loads or hammering a nail, the latter being the topic of this paper. Other impact tasks have been extensively studied: Cisneros et al. [8] focused on impulsive pedipulation on a spherical object with a humanoid robot, van Steen et al. [9] proposed a control framework using Quadratic Programming (QP) to work with simultaneous impacts, Tassi et al. [10] developed a method using velocity matching and a motion planner for catching high-impact flying objects to reduce hardware damage to the robot. Furthermore, Wang and Kheddar [11] worked on robot damage mitigation by designing a conservative, robust and impact-friendly controller using QP. Our overall objective is to hammer a target (a nail in our case) with the maximum impact and mitigate the recoil damage on the robot's joints.

This paper focuses on the impact maximization and the trajectory of the hammer tip to the nail using QP and a 5^{th} order Bézier curve. Section II recalls the robot dynamic model that we use, in Section III we propose a model for the nail-hammer impact, Section IV details the usage of QP in our work to maximize the impact momentum, in Section V we explain how to generate the trajectory of the end-effector to the nail, then we present simulations using the mc_rtc framework [12] in Section VI, and finally we conclude our paper in Section VII.

II. Robot dynamic model

A. Robot dynamic equation

Let n be the number of Degrees Of Freedom (DOF) of the humanoid robot. The dynamic equation of the robot writes:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{u} + \sum_i \mathbf{J_i}(\mathbf{q})^{\mathbf{T}}\mathbf{F_i} \quad (1)$$

where $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \in \mathbb{R}^n$ are the configuration, configuration velocity and configuration acceleration of the robot, respectively, $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n\times n}$ is the symmetric mass matrix or inertia matrix of the robot, $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) \in \mathbb{R}^{n\times n}$ is the dynamic factorization of the Coriolis matrix [13], $\mathbf{g}(\mathbf{q}) \in \mathbb{R}^n$ is the vector of gravitational effects, $\mathbf{u} \in \mathbb{R}^n$ is the input vector of torques, $\mathbf{F_i} \in \mathbb{R}^6$ is the vector of the i-th external wrench (or generalized forces), i.e. forces and torques on each axis, $\mathbf{J_i}(\mathbf{q}) \in \mathbb{R}^{6\times n}$ is the absolute Jacobian of the point of application of $\mathbf{F_i}$, and $\{\cdot\}^{\mathbf{T}}$ is the matrix transpose operator.

B. Jacobian of the end-effector pose

Let $\mathbf{J} \in \mathbb{R}^{6\times n}$ be the Jacobian matrix of the end-effector pose defined such that:

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}} = \begin{bmatrix}\dot{\mathbf{x}}_\nu^{\mathbf{T}} & \dot{\mathbf{x}}_\omega^{\mathbf{T}}\end{bmatrix}^{\mathbf{T}} \quad (2)$$

where $\dot{\mathbf{x}} \in \mathbb{R}^6$ is the task space velocity (translational and angular velocities) of the end-effector, $\dot{\mathbf{x}}_\nu$, $\dot{\mathbf{x}}_\omega \in \mathbb{R}^3$ are the translational and angular part of $\dot{\mathbf{x}}$, respectively.

† CNRS-AIST JRL (Joint Robotics Laboratory), IRL3218, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Japan. Emails: jimmyvu2016@yahoo.fr, ruud.erens@solarteameindhoven.nl, hstefanelli@student.ethz.ch, rafael.cisneros@aist.go.jp, mehdi.benallegue@aist.go.jp, abdelaziz.benallegue@uvsq.fr. * Corresponding author. # Versailles University.

The $\mathbf{J}$ matrix can be computed using forward kinematics and written using two matrices $\mathbf{J}_\nu$, $\mathbf{J}_\omega \in \mathbb{R}^{3\times n}$:

$$\mathbf{J} = \frac{\partial \mathbf{x}}{\partial \mathbf{q}} = \begin{bmatrix} \mathbf{J}_\nu^{\mathbf{T}} & \mathbf{J}_\omega^{\mathbf{T}} \end{bmatrix}^{\mathbf{T}} \tag{3}$$

where $\mathbf{J}_\nu$ is its translational part and $\mathbf{J}_\omega$ is its angular part.

III. Impact modeling

In this section, we propose a mathematical model of the impact between the hammer tip and the nail. This part is heavily inspired by Stronge [14] and his particle impact model.

A. Common tangent plane of collision

In the sense of Stronge [14], we define the common tangent plane of collision as the plane passing through the contact point at the instant of collision that is tangent to both surfaces involved in the collision. The normal unit vector to the collision $\mathbf{n} \in \mathbb{R}^3$ is defined as a vector orthogonal to that plane; see the schematic on Fig. 1.

Let $\mathbf{R}_{w\rightarrow nail} \in SO(3)$ be the rotation matrix of the nail with respect to the world frame w. Then, the normal vector expressed in the world frame $\mathbf{n}_w$ is:

$$\mathbf{n}_w = \mathbf{R}_{w\rightarrow nail}^{\mathbf{T}} \mathbf{n} \tag{4}$$

B. Effective mass of the robot

We define $m_{e,\mathbf{n}}(\mathbf{q}) \in \mathbb{R}^+$ (the "e" stands for "effective"), the configuration dependent Effective Mass of the Robot (EMR), as:

$$m_{e,\mathbf{n}}(\mathbf{q}) \triangleq \left(\mathbf{n}^{\mathbf{T}} \mathbf{\Lambda}(\mathbf{q}) \mathbf{n}\right)^{-1} \tag{5}$$

where $\mathbf{\Lambda} \in \mathbb{R}^{3\times 3}$ is the mobility matrix [8] defined as:

$$\mathbf{\Lambda}(\mathbf{q}) \triangleq \mathbf{J}_\nu(\mathbf{q}) \mathbf{M}^{-1}(\mathbf{q}) \mathbf{J}_\nu^{\mathbf{T}}(\mathbf{q}) \tag{6}$$

The EMR represents the equivalent mass of the robot along the direction of $\mathbf{n}$ if we were to replace the robot by a particle.

C. Choice of the model

We have three main ways to solve the impact mechanics problem: i) a force-based model, ii) an energy-based model and iii) a momentum-based model.

1) Force-based model: The force-based model relies on Newton's second law:

$$\mathbf{F}^h = m_{e,\mathbf{n}} \frac{d\dot{\mathbf{x}}_\nu}{dt} \tag{7}$$

where $\dot{\mathbf{x}}_\nu \in \mathbb{R}^3$ is the translational part of $\dot{\mathbf{x}}$ defined in (2), $\mathbf{F}^h \in \mathbb{R}^3$ is the force applied to the nail by the hammer (the superscript "h" stands for "hammer"). The problem with this model is the discontinuous velocity of the hammer tip at the instant of collision, t_c, as the time derivative at this point in time does not exist. Therefore, we can not use a force-based model, except if we relax the instantanous nature of the impact model which is currently under study.

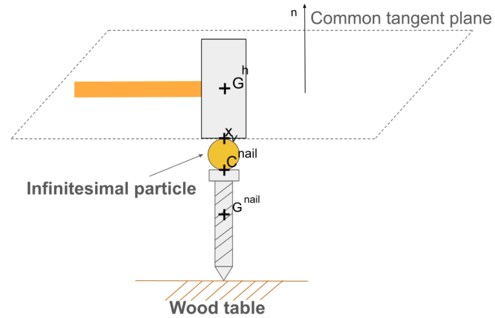

Fig. 1. Schematic of the collision between the hammer tip and the nail modeled as a particle impact.

2) Energy-based model: An energy-based model could be used to overcome the problem of discontinuity of the hammer tip velocity. However, during impact, the kinetic energy of the hammer tip is not fully transmitted to the nail: some of the energy is converted into heat, sound and deformation energy, which are difficult to estimate. Thus, although the model is still usable, we would have to estimate the losses, making the model less precise.

3) Momentum-based model: The momentum based model does not have the problems of the force-based model and the energy-based model. The main advantage of using this model is its ability to deal with discontinuous velocities, which occur when hitting a nail with a hammer. Furthermore, with momentum, the microscopic interactions between the hammer tip and the nail are taken into account through a scalar number e_* called *coefficient of restitution*, which measures the elasticity of the collision. This coefficient can be used to study the post-impact dynamics of the colliding objects; however, this is out of the scope of this article and currently under study. As such, we use the momentum model in the following part of the paper. More precisely, we use a model based on particle impact. A schematic depicting the problem is proposed in Fig. 1. In this model, an infinitesimal particle (i.e. with negligible size) models the local deformations of the hammer tip and the nail.

D. Direct impact model

Let $\mathbf{x}_\nu(t)$ and $\mathbf{c}^{nail}(t) \in \mathbb{R}^3$ be the contact points of the hammer tip and the nail expressed in the world frame w, respectively, as illustrated by Fig. 1. At the instant of contact t_c, these points collide and thus share the same coordinates (neglecting the radius of the particle, see Fig. 1):

$$\mathbf{x}_\nu(t_c) = \mathbf{c}^{nail}(t_c) \tag{8}$$

The velocity and acceleration of the hammer tip are given by the trajectory that will be introduced in Section V-C.

A direct impact is an impact that occurs when the relative approaching velocity of the colliding bodies is parallel to $\mathbf{n}$ [14]. Our goal with the hammering problem is to achieve a direct impact. We will assume that it is indeed the case, thus there will be no tangential sliding during impact.

E. Projected momentum of the hammer tip

To quantify an impact, we define the normal impulse $i \in \mathbb{R}$ given by the hammer tip to the nail over an infinitesimal duration δt to be:

$$i \triangleq \int_{t_c}^{t_c+\delta t} \mathbf{n}^{\mathbf{T}} \mathbf{F}^h dt \tag{9}$$

To maximize the impact, we want to maximize the impulse given by (9) which is equivalent to maximizing the projected momentum $P(t) \in \mathbb{R}$ of the hammer tip along the normal $\mathbf{n}$ [15] defined as:

$$P(t) \triangleq m_{e,\mathbf{n}}(\mathbf{q})\dot{\mathbf{x}}_\nu^{\mathbf{T}}\mathbf{n} \tag{10}$$

This is because:

$$i = \Delta P = P(t_c + \delta t) - P(t_c) \tag{11}$$

For a fixed given target normal velocity $\dot{\mathbf{x}}_\nu^{\mathbf{T}}\mathbf{n}$ (the velocity we want the hammer tip to reach at the nail), we see that in order to maximize the projected momentum of the hammer tip, we have to maximize the EMR. This procedure is explained in Section IV.

IV. Maximization of the effective mass of the robot using QP

The following section presents our QP formulation, in which EMR maximization is included as a weighted objective.

A. Tasks

Let us first introduce what we call a "task". A task is an objective that the robot has to accomplish, for instance maintaining a posture or moving its end-effector to a desired position. To realize a task, we compute a desired configuration acceleration $\ddot{\mathbf{q}}_{\mathbf{d}}$, then we track how well the robot realizes this task using a cost function expressed as a function of the actual configuration acceleration $\ddot{\mathbf{q}}$.

B. Formulation of the problem to solve

For this part, our decision variable is the configuration acceleration $\ddot{\mathbf{q}}$. In order to make it appear at the expression of the effective mass, we differentiate the EMR twice with respect to time:

$$\dot{m}_{e,\mathbf{n}}(\mathbf{q},\dot{\mathbf{q}}) = \frac{dm_{e,\mathbf{n}}}{d\mathbf{q}}\frac{d\mathbf{q}}{dt} = \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\dot{\mathbf{q}} \tag{12}$$

$$\ddot{m}_{e,\mathbf{n}}(\mathbf{q},\dot{\mathbf{q}},\ddot{\mathbf{q}}) = \frac{d}{dt}\left\{\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\right\}\dot{\mathbf{q}} + \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{13}$$

From (13), we can reconstruct $m_{e,\mathbf{n}}$ by discrete time integration. Let Δt be the timestep of the controller, then we have:

$$m_{e,\mathbf{n}}(\mathbf{q}(t+\Delta t)) = m_{e,\mathbf{n}} + \dot{m}_{e,\mathbf{n}}\Delta t + \ddot{m}_{e,\mathbf{n}}\frac{\Delta t^2}{2} \tag{14}$$

Taking $\arg\max_{\ddot{\mathbf{q}}}$ on both sides, the first two terms of the right hand side of the equation are discarded because they do not depend on $\ddot{\mathbf{q}}$, then by (5) and (12) we have:

$$\arg\max_{\ddot{\mathbf{q}}} m_{e,\mathbf{n}}(\mathbf{q}(t+\Delta t)) = \arg\max_{\ddot{\mathbf{q}}} \ddot{m}_{e,\mathbf{n}} \tag{15}$$

Thus, in order to maximize $m_{e,\mathbf{n}}$ with respect to $\ddot{\mathbf{q}}$, we have to maximize $\ddot{m}_{e,\mathbf{n}}$. Therefore, we want to solve:

$$\arg\max_{\ddot{\mathbf{q}}} \left\{ \frac{d}{dt}\left\{\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\right\}\dot{\mathbf{q}} + \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \right\} \tag{16}$$

Again, because our decision variable is $\ddot{\mathbf{q}}$, we can drop the first term which does not depend on the configuration acceleration $\ddot{\mathbf{q}}$ and we are left with:

$$\arg\max_{\ddot{\mathbf{q}}} \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{17}$$

which avoids differentiating the gradient with respect to time. As the QP is defined as a minimization problem, we can rewrite (17) as such in the following way:

$$\arg\min_{\ddot{\mathbf{q}}} -\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{18}$$

This objective function is the Effective Mass Maximization Task (EMMT). The goal of this task is to increase as much as possible the EMR at the next iteration of the controller. However, we cannot assert that the result will be a global maximum because our problem is not necessarily convex. Note that this scheme shares some similarities with the work of Wang and Kheddar [11], where the momentum of the impact is maximized using a QP. However, in that work the effective mass was considered constant and its maximization was not part of the objectives.

C. Linear QP constraints to be respected

Our linear constraints for the QP solver are described as i) joint limits, ii) joint velocity limits, and iii) joint torque limits. These are all defined in the sense of Vaillant et al. [16].

D. Implementation of the EMMT

The posture task is a task where the robot has to maintain a given configuration $\mathbf{q}$. Cisneros et al. [17] gives a deeper look into it. From the complete cost function in our QP formulation:

$$\arg\min_{\ddot{\mathbf{q}}} \frac{1}{2}W_p\|\ddot{\mathbf{q}} + \mathbf{b}\|^2 - W_m\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} + A(\ddot{\mathbf{q}}) \tag{19}$$

the first term represents the posture task, the second term is the EMMT and $A(\ddot{\mathbf{q}}) \in \mathbb{R}$ is the cost function associated to the other possible tasks that the robot has to realize (for instance, a trajectory task). W_p, $W_m \in \mathbb{R}^+$ are the relative weights associated to the posture task and the EMMT, respectively, and $\mathbf{b} \in \mathbb{R}^n$ is a PD controller written as:

$$\mathbf{b} = K_p(\mathbf{q} - \mathbf{q_d}) + K_d(\dot{\mathbf{q}} - \dot{\mathbf{q}}_{\mathbf{d}}) - \ddot{\mathbf{q}}_{\mathbf{d}} \tag{20}$$

where K_p and K_d are the proportional (stiffness) and derivative (damping) gains of the posture task, respectively, $\mathbf{q_d}$, $\dot{\mathbf{q}}_{\mathbf{d}}$, $\ddot{\mathbf{q}}_{\mathbf{d}} \in \mathbb{R}^n$ are the desired configuration, desired configuration velocity and desired configuration acceleration, respectively.

The relationship between K_p and K_d is given by:

$$K_d = 2\sqrt{K_p} \tag{21}$$

To implement our EMMT task, we will modify (19) in the following way:

$$\arg\min_{\ddot{\mathbf{q}}} \frac{1}{2} W_p \|\ddot{\mathbf{q}} + \mathbf{b} - \frac{W_m}{W_p} \nabla m_{e,\mathbf{n}}(\mathbf{q})\|^2 + A(\ddot{\mathbf{q}}) \tag{22}$$

subject to the constraints presented in Section IV-C. This guarantees the problem is strictly convex and can be solved with least square solvers.

This approach allows us to maximize the EMR online instead of optimizing offline. More importantly, it would require high tracking gains to strictly follow the planned kinematic trajectory which increases the risks of robot damage.

V. Trajectory of the end-effector

In this section, we present how the end-effector of the robot reaches the nail with a given velocity.

A. B-Spline trajectory task

The primary task of the end-effector is to reach the nail with a given velocity. To do that, we use a continuous Bézier curve as our trajectory, which is a specific case of a B-Spline. That task is called the B-Spline trajectory task (BSTT).

A $(N+1)$-points Bézier curve $\mathbf{\Gamma} : \mathbb{R} \to \mathbb{R}^3$, or N-th order Bézier curve, is defined as:

$$\mathbf{\Gamma}(t) \triangleq \sum_{k=0}^{N} B_k^N \left(\frac{t - t_i}{T} \right) \mathbf{p}_k \quad \forall t \in [t_i, t_c] \tag{23}$$

where $[t_i, t_c]$ is the time interval on which $\mathbf{\Gamma}$ is defined, $t_c > t_i$, $\mathbf{p}_k \in \mathbb{R}^3$ is the constant k-th control point of the curve, $T = t_c - t_i$ and B_k^N is a Bernstein polynomial that acts as a weight [18], such that:

$$B_k^N \left(\frac{t - t_i}{T} \right) \triangleq \binom{N}{k} \left(\frac{t - t_i}{T} \right)^k \left(1 - \frac{t - t_i}{T} \right)^{N-k} \tag{24}$$

The first control point of the curve, $\mathbf{p_0}$, is the 3D initial position of the hammer tip, $\mathbf{x}_\nu^{t_i} \in \mathbb{R}^3$ (the subscript "*i*" stands for "*initial*"), and the last control point, $\mathbf{p_N}$, is the 3D position of the nail $\mathbf{c}^{nail} \in \mathbb{R}^3$, both with respect to the fixed world frame. Thus:

$$\mathbf{p_0} = \mathbf{x}_\nu^{t_i} \tag{25}$$

$$\mathbf{p_N} = \mathbf{c}^{nail} \tag{26}$$

These points have the property of acting as waypoints for the trajectory, which is not the case for the potential in-between points.

B. Curve constraints

To make the end-effector reach the desired velocity when hitting the nail, we constrain the curve by adding four control points $\mathbf{p_{c_1}}$, $\mathbf{p_{c_2}}$, $\mathbf{p_{c_{N-1}}}$ and $\mathbf{p_{c_{N-2}}} \in \mathbb{R}^3$. These added points count towards the degree of the curve. The expression of the added points is given by [19]:

$$\mathbf{p_{c_1}} = \mathbf{p_0} + \frac{T}{N} \boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_i}} \tag{27}$$

$$\mathbf{p_{c_2}} = 2\mathbf{p_{c_1}} - \mathbf{p_0} + \frac{T^2}{(N-1)(N-2)} \boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_i}} \tag{28}$$

which are added just after $\mathbf{p_0}$, where $\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_i}} \in \mathbb{R}^3$ are the initial linear velocity constraints and $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_i}} \in \mathbb{R}^3$ are the initial linear acceleration constraints. Also:

$$\mathbf{p_{c_{N-1}}} = \mathbf{p_N} - \frac{T}{N-1} \boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_c}} \tag{29}$$

$$\mathbf{p_{c_{N-2}}} = 2\mathbf{p_{c_{N-1}}} - \mathbf{p_N} + \frac{T^2}{(N-1)(N-2)} \boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_c}} \tag{30}$$

where $\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_c}} \in \mathbb{R}^3$ are the linear final velocity constraints and $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_c}} \in \mathbb{R}^3$ are the final linear acceleration constraints. Thus, if we add the curve constraints, we will have six control points in total, hence a 5^{th} order Bézier curve.

C. Hammer tip velocity and acceleration

The desired velocity $\dot{\mathbf{x}}_{\nu,d}$ and acceleration $\ddot{\mathbf{x}}_{\nu,d}$ of the hammer tip from the initial instant t_i to the instant of contact t_c expressed with respect to the world frame w are given by the first and second time derivative of (23), respectively:

$$\dot{\mathbf{x}}_{\nu,d}(t) = \dot{\mathbf{\Gamma}}(t) \quad \forall t \in [t_i, t_c] \tag{31}$$

$$\ddot{\mathbf{x}}_{\nu,d}(t) = \ddot{\mathbf{\Gamma}}(t) \quad \forall t \in [t_i, t_c] \tag{32}$$

Both the velocity and acceleration of the hammer tip are affected by the curve constraints (more precisely, the added control points) defined in Section V-B.

Let $\epsilon_{\mathbf{x}_\nu} = \mathbf{x}_{\nu,d} - \mathbf{x}_\nu$ where $\mathbf{x}_{\nu,d}$ is the desired end-effector position given by $\mathbf{\Gamma}$. The cost function of the BSTT, $A_B(\ddot{\mathbf{q}})$, is:

$$A_B(\ddot{\mathbf{q}}) = W_B \| K_{p,B} \boldsymbol{\epsilon}_{\mathbf{x}_\nu} + K_{d,B} \dot{\boldsymbol{\epsilon}}_{\mathbf{x}_\nu} + \ddot{\mathbf{x}}_{\nu,d} - \dot{\mathbf{J}}_\nu \dot{\mathbf{q}} - \mathbf{J}_\nu \ddot{\mathbf{q}} \|^2 \tag{33}$$

where $W_B \in \mathbb{R}^+$ is the relative weight associated to the BSTT and $K_{p,B}$, $K_{d,B} \in \mathbb{R}^+$ are its stiffness and damping, respectively.

D. Orientation of the hammer tip

In addition to reaching the nail, we have to orient the hammer tip such that the impact with the nail axis is *collinear*. Let $\mathbf{G}^h$, $\mathbf{G}^{nail} \in \mathbb{R}^3$ be the center of mass of the hammer tip and the center of mass of the nail, respectively. A collinear impact is an impact where $\mathbf{G}^h$, $\mathbf{G}^{nail}$ and the contact points $\mathbf{x}_\nu$ and $\mathbf{c}^{nail}$ are aligned [14], see Fig. 1. We constraint two rotation DOFs: the roll and the pitch of the hammer tip, and we let the yaw be free. Indeed, we want to align three vectors: the vector normal to the contact surface of the hammer tip which we will call $\mathbf{n}^h \in \mathbb{R}^3$ (expressed in the hammer tip frame), and the normal vector at the top of the nail.

Let $\mathbf{R}_{w \to h}$ be the rotation matrix describing the rotation of the hammer tip from the world frame, $\boldsymbol{\epsilon} = \mathbf{n}^h - \mathbf{n}$ the error vector between the desired vector ($\mathbf{n}$) and the actual vector ($\mathbf{n}^h$), $\dot{\mathbf{x}}_\omega \in \mathbb{R}^3$ the angular velocity vector of the hammer tip in the world frame given by:

$$\dot{\mathbf{x}}_\omega = -\mathbf{R}_{w \to h} \left[\mathbf{n}^h \right]_\times \mathbf{J}_\omega \dot{\mathbf{q}} \tag{34}$$

where $[\cdot]_\times : \mathbb{R}^3 \to \mathbb{R}^{3 \times 3}$ is the skew-symmetric operator defined, for all $\mathbf{a} = (a_x, a_y, a_z)^\mathbf{T} \in \mathbb{R}^3$, as:

$$[\mathbf{a}]_\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \tag{35}$$

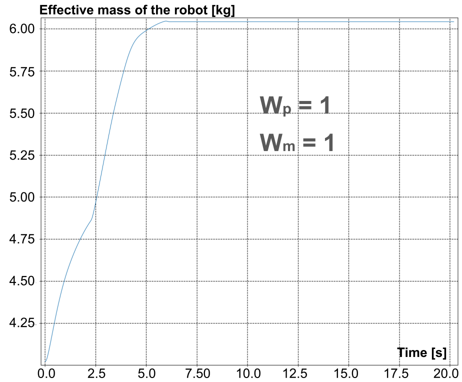

Fig. 2. Time evolution of the EMR with only a posture task active.

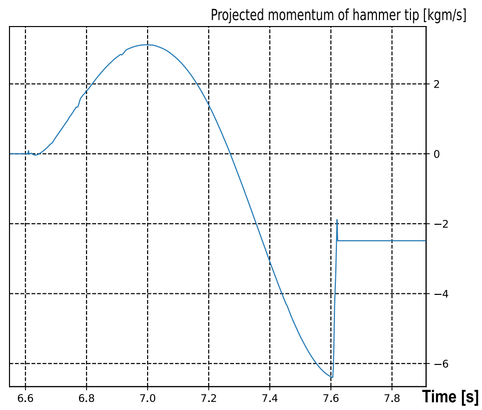

Fig. 3. Time evolution of the projected momentum along the normal $\mathbf{n}$.

The time derivative of (34) is:

$$\ddot{\mathbf{x}}_\omega = -\dot{\mathbf{R}}_{w\to h}\left[\mathbf{n}^h\right]_\times \mathbf{J}_\omega\dot{\mathbf{q}} - \mathbf{R}_{w\to h}\left[\mathbf{n}^h\right]_\times\left(\mathbf{J}_\omega\ddot{\mathbf{q}} + \dot{\mathbf{J}}_\omega\dot{\mathbf{q}}\right) \quad (36)$$

We call this task the Vector Orientation Task (VOT). The cost function of this task is:

$$A_{VOT}(\ddot{\mathbf{q}}) = W_{VOT}\|K_{p,VOT}\boldsymbol{\epsilon} + K_{d,VOT}\dot{\boldsymbol{\epsilon}} - \ddot{\mathbf{x}}_\omega(\ddot{\mathbf{q}})\|^2 \quad (37)$$

where W_{VOT} is the relative weight associated to the VOT and $K_{p,VOT}$, $K_{d,VOT} \in \mathbb{R}^+$ are its stiffness and damping, respectively.

VI. Simulations with the Robot

In this section, we present the simulations made on our humanoid robot HRP-5P. The setting is a fixed hammer attached to one of the robots' grippers (the left one in our case), a fully known position and orientation of the nail and a prefixed desired hitting velocity of the hammer tip on the nail. Simulations were done using mc_mujoco and mc_rtc [20] with its QP solver in torque control mode, and a timestep $\Delta t = 0.002$s, see Fig. 5. The floating base of the robot is fixed as keeping balance after the impact is out of the scope of this paper. The hammer used weighs 0.792 kg, has a length of 328 mm and its contact surface is circular with radius 15.5 mm. The nail is fixed and "floating" in the environment to avoid earlier or post-impact collisions with an object (such as a table), it has a length of 88.03 mm and its head has a radius of 6.95 mm. The tasks at play are the EMMT within the posture task defined in Section IV-B, the BSTT defined in Section V-A and Section V-B, and the VOT defined in Section V-D. The gradient of the effective mass $\nabla m_{e,\mathbf{n}}$ is computed by backward difference. In (20), we set $K_p = 5$, thus by (21) we have $K_d = 4.47$.

The entire simulation goes as follows: i) the robot starts at its half-sitting posture ii) t_i is recorded and the robot starts the hammering motion until an impact is detected on the nail, then t_c is recorded, and finally iii) the robot goes back to its half-sitting posture. An impact is detected on the nail thanks to a force sensor plugin written in ROS2 and added to the nail with mc_rtc: there is an impact if the absolute value of one coordinate of the 3D force applied to the nail is at least greater than some threshold f_e which has been chosen to be 1N to cut off the very small "offset" forces of the simulator, as the magnitude of an impulse is much higher than this value.

We first assess the effectiveness of the EMMT (Section IV) in Fig. 2 with $W_m = 1$ and $W_p = 1$ by isolating the augmented posture task (with EMMT), i.e. we only have the posture task and EMMT active and no other task is active. We see that the EMR is growing over time until it converges to a maximum value. As expected from (22), increasing W_m induces a larger joint acceleration, thus the EMR converges faster and to a larger value; however, it can lead to stability issues with the robot.

Then, we use BSTT and VOT, which are already implemented in mc_rtc and we display the tracking of the theoretical trajectory given by the Bézier curve (23), with the following curve constraints: $\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_i}} = (0,0,0)^{\mathbf{T}}$, $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_i}} = (0,0,0)^{\mathbf{T}}$, $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_c}} = (0,0,0)^{\mathbf{T}}$, and especially

$$\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_c}} = \mathbf{R}^{\mathbf{T}}_{w\to nail}\,{}^{nail}\dot{\mathbf{x}}_\nu^{t_c} \quad (38)$$

where ${}^{nail}\dot{\mathbf{x}}_\nu^{t_c} \in \mathbb{R}^3$ is the input velocity of the hammer tip at the instant of contact t_c expressed in the nail frame. The nail is placed at $\mathbf{c}^{nail} = (0.75, 0.40, 0.90)^{\mathbf{T}}$ from the origin of w with the distances given in meters, and oriented as depicted by Fig. 1, i.e perpendicular to an horizontal plane. We set $T = 1$s, $W_B = 100$, $W_{VOT} = 10$, $W_m = 100$, $W_p = 1$, ${}^{nail}\dot{\mathbf{x}}_\nu^{t_c} = (0, 0, -2.0)^{\mathbf{T}}$ m/s, the stiffness of the BSTT to $K_{p,B} = 100000$ and the stiffness of the VOT to $K_{p,VOT} = 250$ with their damping given by (21). We see the tracking of the hammer tip velocity in Fig. 4(a) and (b). It is worth to mention that the hammer tip properly reaches ${}^{nail}\dot{\mathbf{x}}_\nu^{t_c}$. For $K_{p,VOT} = 250$, we record an orientation error of 11.88 deg and some oscillations at the beginning. The stiffness of the VOT is decreased to 100 in Fig. 4(b), which removed the oscillations. We also tested $K_{p,VOT} = 1000$: the final orientation error has been recorded to be 4.16 deg, however we observed large oscillations in the first instants. Increasing $K_{p,VOT}$ decreases the error between the desired and actual final hammer tip orientations at the cost of larger oscillations at the beginning. The projected momentum of the hammer tip is computed using (10) and is shown in Fig. 3, with the impulse displayed with the sudden jump in projected momentum.

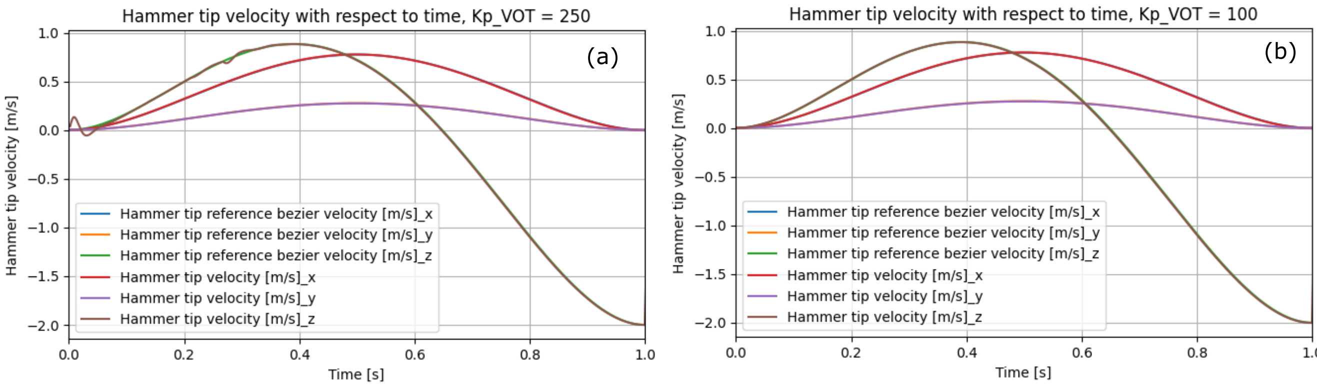

Fig. 4. Velocity tracking of the hammer tip for different $K_{p,VOT}$ values.

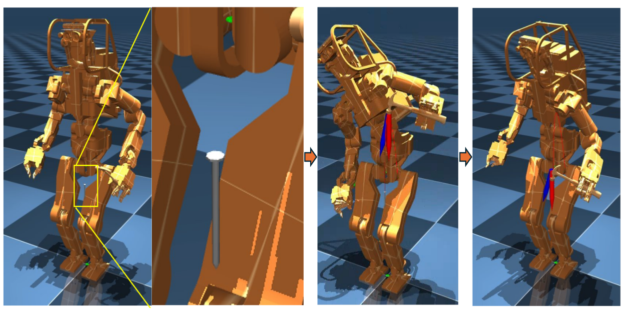

Fig. 5. Snapshots of the simulation in mc_mujoco

VII. Conclusion

In this work, we presented a way to implement a hammering task on a humanoid robot. We proposed our method to maximize the EMR using its gradient and QP, then we suggested an end-effector trajectory built from a constrained 5^{th} order Bézier curve. Simulations showed that the EMR is getting maximized and the trajectory given by the Bézier curve is properly followed by the end-effector.

Possible improvements include improvements on modeling errors in the simulator, an estimation of the nail penetration in an object (like a wood table) along with a simulation of the dynamics of the nail, a way to build our path and orientation to the nail by efficiently considering collision constraints with the environment or a post-impact reference that utilizes the post-impact velocity of the hammer tip to prepare for the next hit as the robot only goes back to its initial posture for now, and experiments on the actual robot.

In the future, we want to work on the stability problem of the robot potentially induced by the post-impact rebounce of the hammer, damage minimization on the robot's joints, feasibility of the hammering task on the free-standing robot and detection of the nail via computer vision.

Acknowledgment

This research was funded by JSPS KAKENHI, grant number 24K00836.

References

[1] Scott P. Schneider. "Musculoskeletal Injuries in Construction: A Review of the Literature". In: *Applied Occupational and Environmental Hygiene* 16.11 (Nov. 2001), pp. 1056–1064.

[2] Julie N. Côté et al. "Differences in Multi-Joint Kinematic Patterns of Repetitive Hammering in Healthy, Fatigued and Shoulder-Injured Individuals". In: *Clinical Biomechanics* 20.6 (July 2005), pp. 581–590.

[3] Nilanthy Balendra and Joseph E. Langenderfer. "Effect of Hammer Mass on Upper Extremity Joint Moments". In: *Applied Ergonomics* 60 (Apr. 2017), pp. 231–239.

[4] Tadej Petrič et al. "Hammering Does Not Fit Fitts' Law". In: *Frontiers in Computational Neuroscience* 11 (May 2017).

[5] F. Aghazadeh and A. Mital. "Injuries Due to Handtools: Results of a Questionnaire". In: *Applied Ergonomics* 18.4 (Dec. 1987), pp. 273–278.

[6] Yuan-Ho Chen and Yi-Lang Chen. "An Optimal Effective Working Height for Hammering Tasks". In: *Work* 61.2 (Nov. 2018), pp. 181–187.

[7] Kenji Kaneko et al. "Humanoid Robot HRP-5P: An Electrically Actuated Humanoid Robot With High-Power and Wide-Range Joints". In: *IEEE Robotics and Automation Letters* 4.2 (Apr. 2019), pp. 1431–1438.

[8] Rafael Cisneros, Kazuhito Yokoi, and Eiichi Yoshida. "Impulsive Pedipulation of a Spherical Object with 3D Goal Position by a Humanoid Robot: A 3D Targeted Kicking Motion Generator". In: *International Journal of Humanoid Robotics* 13.02 (June 2016), p. 1650003. (Visited on 11/01/2025).

[9] Jari J. van Steen, Nathan van de Wouw, and Alessandro Saccon. "Robot Control for Simultaneous Impact Tasks via Quadratic Programming-based Reference Spreading". In: *2022 American Control Conference (ACC)*. June 2022, pp. 3865–3872. arXiv: 2111.05211 [cs].

[10] Francesco Tassi et al. "IMA-catcher: An IMpact-aware Nonprehensile Catching Framework Based on Combined Optimization and Learning". In: *The International Journal of Robotics Research* (June 2025), p. 02783649251345851.

[11] Yuquan Wang and Abderrahmane Kheddar. "Impact-Friendly Robust Control Design with Task-Space Quadratic Optimization". In: *Robotics: Science and Systems XV*. Robotics: Science and Systems Foundation, June 2019.

[12] The mc_rtc framework. URL: https://jrl.cnrs.fr/mc_rtc/.

[13] Magnus Bjerkeng and Kristin Y. Pettersen. "A New Coriolis Matrix Factorization". In: *2012 IEEE International Conference on Robotics and Automation*. May 2012, pp. 4974–4979.

[14] W. J. Stronge. *Impact Mechanics*. 2nd ed. Cambridge University Press, 2018.

[15] Luciano da Fontoura Costa. Single Point Particle Motion: Mass, Force, Momenta, Impulse, Work. 2022. ⟨hal-03907279⟩.

[16] Joris Vaillant et al. "Multi-Contact Vertical Ladder Climbing with an HRP-2 Humanoid". In: *Autonomous Robots* 40.3 (Mar. 2016), pp. 561–580.

[17] Rafael Cisneros et al. "Robust Humanoid Control Using a QP Solver with Integral Gains". In: *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Madrid: IEEE, Oct. 2018, pp. 7472–7479.

[18] David Salomon. *Computer Graphics & Geometric Modeling*.

[19] The ndcurves library. URL: https://github.com/loco-3d/ndcurves/blob/devel/doc/curves.pdf.

[20] Rohan P. Singh, Pierre Gergondet, and Fumio Kanehiro. *Mc-Mujoco: Simulating Articulated Robots with FSM Controllers in MuJoCo*. Sept. 2022. arXiv: 2209.00274 [cs].